Ajay Gunalan

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Full-Stack Roboticist
Learning whatever the robot demands to

A **Full-Stack Roboticist** who loves to build software for robots involving embedded systems, controls, planning, and machine learning. My ultimate goal is to make a dexterous, agile, versatile and robust, robots to solve real-world problems. To achieve that, I think we should have a broad interdisciplinary approach to debug the failures accurately.

Software Experience

Indian Institute of Science, Bangalore, India — *February 2018 – Present*
Project Assistant — Robert Bosch Centre for Cyber-Physical System

I'm involved in developing a **quadruped robot** called *stoch*. We used a Deep Reinforcement Learning to generate the end point trajectory and realized on hardware using Central Pattern Generator. [link]

- Straight line trajectory tracking using IMU.
- Sensors and actuators, interface with linux based single board computers.
- Inter Process Communication between drivers in C and Python using shared memory enabling low latency.
- Non-Blocking I/O for controlling the robot during teleoperation.
- CAN bus communication between two debian based system.
- Real-Time Embedded Programming on STM32F4 board using ChibiOS (Basics).
- A thorough understanding of servos (inner-working), actuator design (Quasi Direct Drive. 4Q control, compliance) and overall mechtronic system architecture through literature survey.

ASIMOV Robotics Pvt. Ltd, Kochi, India — *July 2017 – December 2017*
Intern — Software Development (R&D) Division

We developed a **banking service robot** and arm manipulators for specific clients. [link]

- Gravity compensation of arms for assistive mode.
- Position and Velocity control of DC motor in Arduino using PID.
- TCP/IP communication between ROS and non-ROS module.
- Motion planning simulation in Gazebo using ROS and MoveIt.
- Sensor Integration such as IMU, Tactile (FSR) and Ultrasonic.

Education

B.S. Abdur Rahman University, Chennai, India — *August 2013 – May 2017*
Bachelors of Technology in Mechanical Engineering — CGPA: 8.45 / 10.00

GRT Mahalakshmi Vidyalaya Matriculation Higher Secondary School, Chennai, India — *March 2013*
12th Grade, Tamil Nadu State Board — Percentage: 88.00 %

A.V. Meiyappan Matriculation Higher Secondary School, Chennai, India — *March 2011*
10th Grade, Tamil Nadu Matriculation Board — Percentage: 87.40 %

Research Interest

- Combining Machine Learning and Motion Planning.
- Manipulation of Deformable Objects.
- Grasping, Tactile Sensing, Contact-Dynamics.
- Learning to Control (Model-Based RL, Imitation Learning and Meta-Learning).
- Application of Non-Linear Control and Optimization Techniques in Real-Time on robots.
- Medical Robots, Hybrid-Systems, Bio-Inspired Robotics.

Projects

- Actuators for dexterous and agile robots. [link]
  - A robot is a bunch of actuators connected. Thus, it is of prime importance to exactly understand the requirements on the hardware to build a dexterous, agile and robust robot.
- Smart Intravenous Dripper. [link]
  - Automation of Intravenous drippers to prevent air-embolism and reversal of blood-flow in the patients.
- Open-Source BLDC driver with CAN communication. [link]
  - An attempt to make customized BLDC driver based upon open source hardware which resulted in a huge learning for me.

Publication(s)

[Google Scholar], [ORCID ID], [arXiv identifier]

- Dholakiya D., Bhattacharya S., **Ajay G.**, Singla A., Shalabh B., Bharatwaj A., Ghosal A., Shishir K., "*Design, Development and Experimental Realization of a Quadrupedal Research Platform: Stoch*" In Proceedings of 2019 IEEE International Conference on Control, Automation and Robotics (ICCAR 2019), Beijing, China. [video] [arXiv]

Technical skills

Programming	C, C++, Python, Html, MATLAB
Framework	ROS, MoveIt, TensorFlow, POSIX
OS	Ubuntu, Windows, ChibiOS (Embedded RTOS)
Simulation	Gazebo, ANSYS Workbench
Version & Build	Git, GitHub, Make, CMake, Catkin, Jekyll
Debugging Tools	OpenOCD, gdb, Valgrind, strace, ptrace
Single Board Computer	Raspberry-Pi, Odroid, Beaglebone
Microcontroller	STM32F4 series (ARM Cortex-M4), Arduino
Bus & Network	CAN, I^2C, SPI, Serial, Socket (TCP/IP), MQTT
Servos & Motors	Dynamixel, Kondos, pwm enabled servos, DC & stepper motors
Others	Doxygen, LaTeX, pandoc, markdown, MS-Office, Prezi, Solidworks

Honors and Awards

- **Finalist, Top 10 out of 11,000+ applicants, in India Innovation Challenge Design Contest 2016** conducted by Department of Science and Technology, Government of India & Texas Instruments Inc., and anchored by Indian Institute of Management, Bangalore for our project, "Smart Intravenous Dripper".
- ISHRAE $2016 - 2017$ President in B.S. Abdur Rahman University and was responsible for organizing and conducting various events related to ISHRAE.
- Won 1st Prize in ISHRAE Chennai chapter Level Quiz by St. Joseph's College of Engineering, Chennai under the auspices of ISHRAE.

- Won 1st Prize in Technical Quiz at Confluence 2016, A National Level Technical Symposium.
- Finalist in the Semifinals of AQuest, An All India Inter-Collegiate Engineering Quiz Contest at Kolkata by ISHRAE.
- Won 1st Prize in Design of Heat Exchanger at Mechfest 2016, Technical Symposium.
- Won 1st Prize in Tamil Oration at A.V. Meiyappan Matriculation Higher Secondary School.
- Won 3rd Prize in Choreography at Festeve 2014 by Women's Christian College, Chennai.
- Won 3rd Prize in Burn the floor at Chrysalis 2014 by Loyola Institute of Business Administration, Chennai.
  Won 3rd Prize in Western Dance at Srishti 2014 by Ethiraj Collage for Women, Chennai.

Miscellaneous

- Took seminars on simulation of a robotic arm in Gazebo using ROS and MoveIt for students of Dr. Shishir Kolathaya and Prof. Ashitav Goshal.
- Internship (July 2016) at TIDC INDIA, Ambattur, India, where I learnt various process and methodologies involved in desgin and fabrication of cam chain used in two-wheelers.
- Internship (June 2016) at J.K. Fenner(India) Ltd, Sriperumbudur, India, where I learnt various process and methodologies involved in desgin and fabrication of rubber seal's used in bearings.
- In-plant Training (June 2015) at Ashok Leyland, Ennore, India, where I had a practical exposure to various manufacturing methods and assemble line production system.
- During my bachelors, I actively participated in various conference and workshops to enhance my knowledge across the technical domain.

References

- Prof. Bharadwaj Amrutur, Chairman, Robert Bosch Center for Cyber-Physical Systems, Indian Institute of Science, Bangalore, India.
  Email: amrutur@iisc.ac.in
- Dr. Prince Arockia Doss, Professor, Mechanical Department, B.S. Abdur Rahman Crescent Institute of Science and Technology, Chennai, India.
  Email: princeadoss@crescent.education
- Mr. Nirmal Thomas, Technical Team Lead, ASIMOV Robotics Pvt. Ltd, Kochi, India.
  Email: nirmalthomas@asimovrobotics.com